

## CS620 – Modelling and Simulation

### Printable Midterm Prep (Topics 1–50)

#### ◆ Topic-wise Quick Notes (1–50)

- 1. Simulation:** Simulation is the imitation of the operation of a real-world system or process to study its behavior under different scenarios.
- 2. When Simulation is Appropriate:** Used when experimentation with the real system is difficult, dangerous, or expensive.
- 3. When Simulation is Not Appropriate:** Not suitable when simple analytical solutions or direct calculations are possible.
- 4. Advantages of Simulation:** Helps analyze complex systems, test 'what-if' scenarios, and train people without real-world risks.
- 5. Disadvantages of Simulation:** Can be expensive, time-consuming, and results are only as good as the model's accuracy.
- 6. Application Areas:** Manufacturing, healthcare, military, supply chain, traffic systems, business processes.
- 7. Military Applications:** Used for training, logistics, battle planning, and weapon system evaluation.
- 8. System & Environment:** System: focus area; Environment: external factors affecting the system.
- 9. System Concepts:** Includes homeostasis, adaptation, feedback loop, throughput, etc.
- 10. Components of a System:** Entity, Attribute, State, Activity, Event, and Delay.
- 11. Discrete vs Continuous:** Discrete: changes at specific times (bank queue). Continuous: smooth change (water level in tank).
- 12. Model of a System:** A simplified representation of a real-world system to study it.
- 13. Types of Models:** Static/Dynamic, Deterministic/Stochastic, Discrete/Continuous.
- 14. Discrete Event Simulation:** Simulation where state changes at discrete points using event list.
- 15–22. Steps of Simulation Study:** Planning → Model → Data → Verification → Experiment → Reporting → Implementation.
- 23. Simulation Case Study Steps:** Define goal → Build model → Validate → Run → Analyze.

**24–27. Spreadsheet Simulations:** Use RAND(), simulate randomness (coin toss, arrivals, service time).

**28–29. Simulation Table:** Organize input, random data, logic, and result columns in table form.

**30–34. Discrete Event Framework:** Defines Clock, Events, Activities, State, and Entity interactions.

**35–38. Event Scheduling Algorithm:** 3 phases: A (remove event), B (execute), C (trigger new events).

**39–40. World Views:** Three views: Event Scheduling, Activity Scanning, Process Interaction.

**41–43. Manual Simulation:** Simulation done step-by-step using paper or spreadsheet.

**44. Statistical Models:** Use distributions (Poisson, Exponential, Normal) to model randomness.

**45–48. Review Terms:** Entity, Attribute, State, Activity, Event, Delay, Clock, System, Environment.

**49–50. Useful Models:** Queueing and inventory simulation models used in business and industry.

## Printable Midterm Prep (Topics 51–100)

### ◆ Topic-wise Quick Notes (51–100)

**51–55. Queuing Models:** Used to simulate lines of people, items, or requests waiting for service. Includes arrival rate, service rate, queue discipline.

**56–60. Inventory Systems:** Simulate ordering, stocking, and selling products. Helps find optimal stock level and reorder policies.

**61–63. Delay in Simulation:** Delays occur when entities must wait for a resource. Modeled using probability or event dependencies.

**64–66. Activity vs. Event:** Activity: takes time (e.g., service). Event: instantaneous change (e.g., arrival, departure).

**67–70. Simulation Clock:** Keeps track of simulation time. It jumps to next event time in discrete event simulation.

**71–73. Entity Attributes:** Attributes define properties like arrival time, priority, or service type for each entity.

**74–76. Future Event List (FEL):** A time-ordered list of all scheduled events in the system. Drives simulation progress.

**77–79. Random Numbers:** Use uniform distributions (0–1) with functions like RAND(). Convert to specific distributions for modeling.

**80–83. Probability Distributions:** Exponential, Poisson, Normal – used to simulate arrival, service, and failure times.

**84–85. Validation Techniques:** Compare model results with real-world data, use expert opinion, or run statistical tests.

**86–88. Verification Techniques:** Ensure model logic is implemented correctly. Use walkthroughs, trace events, and code checks.

**89–91. Animation in Simulation:** Visualization of system behavior (e.g., moving cars, customer flow). Enhances understanding and debugging.

**92–94. Statistical Output Analysis:** Analyzing results from multiple runs: mean, variance, confidence intervals.

**95–96. Terminating vs Non-Terminating Systems:** Terminating: simulation ends (e.g., 8-hour shift). Non-Terminating: runs indefinitely (e.g., ATM).

**97–98. Warm-up Period:** Ignore initial output to allow the system to reach a stable state. Helps avoid bias.

**99–100. Use of NetLogo:** NetLogo is a tool for agent-based simulation. Used for modeling decentralized systems (e.g., ant colonies, fire spread).

## Topic-wise Notes (101–125)

### 101–103. Agent-Based Modeling (ABM)

ABM is a simulation method that models the behavior of individual agents (like people, animals, or machines) that act independently and interact with each other to create system-level behavior.

### 104–106. Characteristics of ABM

- Agents act autonomously
- Local rules, no central control
- Agents may adapt, learn, and evolve
- Collective (emergent) behavior arises from interactions

### 107–108. Benefits of ABM

- Models individual behaviors
- Captures complex interactions
- Visual and flexible
- Useful when system behavior is not easily predictable

### 109–110. Applications of ABM

Used in:

- Biology (cells, immune systems)
- Sociology (crowd behavior)

- Traffic simulation
- Marketing and economics
- Epidemic modeling

#### **111–112. Fire Model in NetLogo**

Simulates fire spread in a forest where trees catch fire and pass it to adjacent trees. Useful to study spread rate and control methods.

#### **113–114. Ant Model in NetLogo**

Models ants searching for food using pheromones. Demonstrates **swarm intelligence** — where simple agents (ants) create complex behavior.

#### **115–117. Model Validation in ABM**

Compare simulation results to real-world data or expected patterns. Experts can validate if the model behavior is realistic.

#### **118–119. Advantages of NetLogo**

- Easy to use
- Ideal for teaching and prototyping
- Visual simulation interface
- Good for decentralized models

#### **120–121. Setting Up ABM in NetLogo**

Define agents, environment, behaviors (rules), interaction logic, and visual output. Then run and observe system behavior.

#### **122–125. Final Summary & Revision**

Summarizes all previous concepts including discrete-event models, spreadsheets, random models, agent-based simulation, applications, and tools.

### **Important MCQs (Topics 1–50)**

1. Simulation is:

- A game
- A real system
- The imitation of a real system
- A type of database

✓ Correct Answer: The imitation of a real system

2. Which is NOT a reason to use simulation?

- Real system is complex

- Analytical solution is available
- Cost is high
- System is risky

✓ Correct Answer: Analytical solution is available

3. Which of the following is a disadvantage of simulation?

- Gives insights
- Saves cost
- Time-consuming
- Easy to apply

✓ Correct Answer: Time-consuming

4. What is an entity in simulation?

- A property
- An object of interest
- A delay
- A model

✓ Correct Answer: An object of interest

5. Which model uses event list and state changes at points in time?

- Static Model
- Spreadsheet Model
- Discrete Event Simulation
- Physical Model

✓ Correct Answer: Discrete Event Simulation

6. A dynamic model is one which:

- Never changes
- Represents time-based changes
- Only for small systems
- Has no randomness

✓ Correct Answer: Represents time-based changes

7. The RAND() function in Excel generates:

- Negative numbers
- Text values
- Random number between 0 and 1
- Only 0 or 1

✓ Correct Answer: Random number between 0 and 1

8. In a system, delay is:

- A model
- An input
- A known activity
- An unknown time duration

✓ Correct Answer: An unknown time duration

9. Which of the following is NOT a type of simulation model?

- Deterministic
- Dynamic
- Chaotic
- Stochastic

✓ Correct Answer: Chaotic

10. Which one is NOT a simulation step?

- Problem formulation
- Model building
- Cooking
- Validation

✓ Correct Answer: Cooking

### ◆ Important MCQs (Topics 51–100)

1. What is a queuing model used for?

- Stock management

- Process scheduling
- Customer waiting line simulation
- Data encryption

✓ Correct Answer: Customer waiting line simulation

2. Which distribution is used for modeling time between arrivals?

- Normal
- Poisson
- Exponential
- Binomial

✓ Correct Answer: Exponential

3. Future Event List (FEL) is:

- A record of past events
- A to-do list
- List of scheduled future events
- Part of model input

✓ Correct Answer: List of scheduled future events

4. Simulation clock is used to:

- Track real time
- Track simulation time
- Count events
- Measure resources

✓ Correct Answer: Track simulation time

5. Which is an attribute of an entity?

- Queue
- Arrival time
- Server
- Resource

✓ Correct Answer: Arrival time

6. Which is a verification method?

- Compare with real-world
- Statistical analysis
- Code walkthrough
- Expert review

✓ Correct Answer: Code walkthrough

7. Which tool is used for agent-based simulation?

- MS Excel
- Simulink
- NetLogo
- C++

✓ Correct Answer: NetLogo

8. What is warm-up period used for?

- Heating model data
- Improving accuracy
- Ignoring unstable initial data
- Clearing cache

✓ Correct Answer: Ignoring unstable initial data

9. Terminating system example?

- 24/7 ATM
- Hospital emergency room
- 8-hour shift at factory
- Cloud server

✓ Correct Answer: 8-hour shift at factory

10. Which analysis is done on simulation output?

- SQL Query

- Output debugging
- Statistical analysis
- Graph drawing

✓ Correct Answer: Statistical analysis

### ✓ Important MCQs (Topics 101–125)

- What is the key concept of Agent-Based Modeling (ABM)?**
  - A) Centralized control
  - B) Top-down simulation
  - C) Autonomous agents ✓
  - D) Mathematical equations
- Which tool is commonly used for ABM?**
  - A) Simulink
  - B) Excel
  - C) NetLogo ✓
  - D) Arena
- Which of the following is a feature of ABM?**
  - A) Global rules only
  - B) Agents act randomly
  - C) Agents act independently ✓
  - D) Continuous input
- The fire model in NetLogo demonstrates:**
  - A) Economics
  - B) Traffic signals
  - C) Forest fire spread ✓
  - D) Disease outbreak
- What behavior is modeled in the Ant model?**
  - A) Marketing
  - B) Food search using pheromones ✓
  - C) Resource allocation
  - D) Cloud computing
- ABM is especially good for modeling:**
  - A) Simple systems
  - B) Static data
  - C) Emergent behavior ✓
  - D) Single variable systems
- Which of the following is NOT an ABM advantage?**
  - A) Visual output
  - B) Easy to use
  - C) Requires no logic ✓
  - D) Flexible modeling
- In ABM, agents:**
  - A) Follow top-down rules
  - B) Are passive
  - C) Interact based on rules ✓
  - D) Depend on global controller
- NetLogo is best suited for:**

- A) Server programming
- B) Numerical analysis
- C) Agent-based simulation ✓
- D) Cryptography

10. Which is an example of ABM application?

- A) Mathematical optimization
- B) Network protocol design
- C) Crowd behavior simulation ✓
- D) Hardware design

### ◆ Important 3 and 5 Mark Questions (with Answers)

#### 📖 3 Marks Questions

1. Q: Define simulation with example.

Ans: Simulation is the imitation of the operation of a real-world system to study behavior.  
Example: Simulating traffic flow in a city.

2. Q: Differentiate between Discrete and Continuous systems.

Ans: Discrete systems change at specific times (e.g., customer arrivals), while continuous systems change smoothly over time (e.g., water level).

3. Q: List components of a system in simulation.

Ans: Entity, Attribute, State, Event, Activity, Delay – all define how a system behaves in a simulation.

#### 📖 5 Marks Questions

4. Q: Describe the main steps of a simulation study.

Ans: The steps include: (1) Problem Formulation, (2) Setting Objectives, (3) Model Conceptualization, (4) Data Collection, (5) Model Translation, (6) Verification, (7) Validation, (8) Experimentation, (9) Documentation, and (10) Implementation.

5. Q: Explain spreadsheet simulation using an example.

Ans: Spreadsheet simulation uses functions like RAND() to simulate random events. For example, =IF(RAND()<=0.5,"H","T") simulates a coin toss.

6. Q: What is Event Scheduling Algorithm? Describe its phases.

Ans: It is a 3-phase simulation approach: Phase A: Remove imminent event from FEL and update clock. Phase B: Execute the event and update system state. Phase C: Schedule new events if triggered.

### ◆ Important 3 and 5 Mark Questions (with Answers)

#### 📖 3 Marks Questions

7. Q: Define Future Event List (FEL).

Ans: FEL is a time-ordered list containing all upcoming events in the simulation. It ensures that events are processed in correct sequence.

8. Q: Differentiate between Activity and Event.

Ans: Activity takes time (e.g., service), while Event is an instantaneous occurrence (e.g., arrival).

9. Q: What is warm-up period in simulation?

Ans: The time at the beginning of simulation where data may be unstable, and hence ignored for analysis.

#### 5 Marks Questions

10. Q: Explain the use of statistical distributions in simulation.

Ans: Distributions help simulate uncertainty. Exponential is used for inter-arrival time, Poisson for count of events, Normal for service or failure times. They help model real-world randomness.

11. Q: Describe how NetLogo is used for agent-based simulation.

Ans: NetLogo models decentralized systems. Agents (e.g., turtles) follow rules and interact with environment. Used for modeling traffic, fire spread, epidemics, etc.

12. Q: Differentiate between Verification and Validation with examples.

Ans: Verification: Ensuring model is built correctly (e.g., checking code logic). Validation: Ensuring model accurately represents the real system (e.g., comparing results with real data).

### 3-Mark Questions with Answers

**Q1: Define Agent-Based Modeling (ABM).**

**Ans:** ABM is a simulation technique where individual entities (agents) operate independently and interact with others based on rules. Their combined behavior forms the system's overall behavior.

---

**Q2: What is the fire model in NetLogo?**

**Ans:** It simulates how fire spreads through a forest. Each tree can ignite its neighboring trees, helping to visualize the dynamics of wildfire spread.

---

**Q3: List two characteristics of agents in ABM.**

**Ans:**

1. Agents act independently using local rules.
  2. They can interact with other agents and adapt based on experiences.
-

## 5-Mark Questions with Answers

**Q1: Explain the working of the Ant Model in NetLogo.**

**Ans:** The Ant model shows ants leaving their nest to find food. When food is found, ants leave a pheromone trail back to the nest. Other ants follow stronger trails, leading to swarm behavior.

This models decentralized decision-making and communication.

---

**Q2: Discuss advantages and applications of Agent-Based Modeling.**

**Ans:**

**Advantages:**

- Visual and easy to understand
- Captures individual-level behavior
- Can model complex, adaptive systems

**Applications:**

- Crowd simulations
  - Disease spread
  - Ecology (animal movement)
  - Market behavior
- 

**Q3: How can an ABM model be validated?**

**Ans:**

- Compare simulation output with real-world data
- Use expert opinions to check logic and behavior
- Ensure the model replicates known or expected patterns